Fractal Entropic Geometrodynamics: Emergent Gravity and Three Particle Generations from Discrete Topological Constraints

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(Dated: February 18, 2026)

We present a strictly discrete framework in which General Relativity and the three-generation structure of the Standard Model emerge from two axioms and one principle. An $O(N)$ simulation engine—implemented in Rust and freely available—Poisson-sprinkles events into a 4D causal diamond, builds the Hasse diagram, and detects topological defects called *Causal Prisms* ($K_{2,n}$ bipartite subgraphs). A discrete phase invariant based on the sign trichotomy of bulk momentum classifies prisms into exactly three generations ($g \leq 3$), producing a mass hierarchy Gen 1 : Gen 2 : Gen 3 = 4.55 : 6.53 : 7.73 and CPT symmetry—from pure combinatorics. The spectral dimension flows from $d_S(\sigma{=}1) = 1.95 \pm 0.002$ in the UV, crossing $d_S = 4$ at $\sigma \approx 3$–4. A finite-size scaling analysis across five lattice sizes ($N = 10^5$ to 10^7, $M \geq 8$ realisations each) reveals that all boundary-sensitive observables follow the 4D scaling law $\mathcal{O}(N) = \mathcal{O}_\infty + a\,N^{-1/4}$ ($R^2 = 0.9996$ for $\mathcal{Q}_{\rm topo}$), while topological invariants—mass ratios and generation fractions—are N-independent. The emergent fine structure constant $\alpha = \mathcal{Q}_{\rm topo}/(8\pi)$ and self-energy ratio $\Omega = 1/\mathcal{Q}_{\rm topo} - 1$ are locked by the exact identity $\alpha(1 + \Omega) = 1/(8\pi)$; the bare topological coupling $\alpha_0^{-1} = 165.1 \pm 1.0$ sets the UV-cutoff value for renormalization-group studies. Gravity is the Fisher information cost of macroscopic deformation; the Einstein equations are thermodynamic. Bounded Hasse degree ($D \leq 15$) makes the computation $O(N)$, providing a proof—not an assertion—that the physics is local.

I. WHAT THE COMPUTATION PRODUCES

We begin with the output. Theory will follow.

An ensemble of $M = 20$ realisations (seeds 42–61, $N = 10^7$ Poisson-sprinkled events each, converged to $\varepsilon < 0.001$) produces the numbers in Table I. No free parameters were adjusted. No continuous fields were integrated. No gauge groups were imposed. The input is a random number generator and an integer N.

TABLE I. Observables at $N = 10^7$. Ensemble of $M = 20$ realisations (seeds 42–61, converged to $\varepsilon < 0.001$). All quantities are dimensionless or in topological Planck units. Continuum-limit extrapolations via finite-size scaling appear in Table II.

Observable	**Value**
Spectral dimension (σ-resolved)	
d_S vacuum (σ=1, UV)	1.949 ± 0.002
d_S vacuum (σ=4, IR)	4.956 ± 0.001
d_S defect (peak, $\sigma\approx$7)	9.06
Topological defects (per realisation)	
Total prisms	386,197
g=1 / g=2 / g=3	(2.1% / 84.9% / 13.0%)
Mass and symmetry	
Mass: Gen 1 / Gen 2 / Gen 3	4.555 / 6.531 / 7.732
Ratio $m_1 : m_2 : m_3$	1 : 1.43 : 1.70
CPT: Gen 1 vs Anti-1	4.555 vs 4.45 ($\Delta m/m = 2.3\%$)
Emergent coupling constants	
$\mathcal{Q}_{\rm topo} = \sum \lvert\Phi\rvert^2 / \sum N^2$	0.1907
$\alpha = \mathcal{Q}_{\rm topo}/(8\pi)$	1/131.8
$\Omega_{\rm energy} = 1/\mathcal{Q}_{\rm topo} - 1$	4.24
$\alpha\,(1 + \Omega_{\rm energy})$	$1/(8\pi)$ *(exact)*

These numbers require interpretation. Here is what they mean, stated plainly; the derivations occupy the rest of the paper.

Three generations. The engine finds ∼386,000 topological defects (Causal Prisms) per realisation in the Hasse diagram. Each is classified by a discrete invariant $g \in \{1, 2, 3\}$—the number of distinct sign values of a bulk momentum field restricted to the defect. The bound $g \leq 3$ is not a fit: the sign function has three values (−, 0, +), so three is the maximum. This is a theorem (Sec. VI A).

Mass hierarchy. Higher-generation defects trap random walkers more effectively: $m_1 < m_2 < m_3$. Mass is not a coupling constant—it is *topological probability retention*: the enhanced return probability of a walker diffusing through a structured subgraph.

CPT symmetry. Matter and antimatter prisms have equal masses to within 2.3% at M=20 realisations (Gen 1 mass 4.555 vs Anti-1 mass 4.45). Finite-size scaling (Sec. X) extrapolates $\Delta m/m \to 1.2\%$ in the thermodynamic limit: CPT is an exact symmetry of the causal order (Sec. VI B).

Spectral dimension flow. At $\sigma = 1$ (UV), $d_S = 1.949 \pm 0.002$: dimensional reduction consistent with the $d_S \approx 2$ Planck-scale prediction of CDT [1] and causal set results [2, 3]. By $\sigma = 4$ the vacuum d_S crosses 4.0 and reaches 4.956 ± 0.001: the 4D continuum has emerged (Fig. 5). At the defect core, d_S peaks at 9.06 ($\sigma \approx 7$): Causal Prisms trap walkers, inflating the local return probability beyond what a 4D vacuum would produce.

$O(N)$ scaling. Ten million events in 24 minutes. This is not an engineering achievement—it is a *physical statement*: the Hasse degree is bounded ($D \leq 15$), so every operation in the pipeline is local. The simulation's efficiency is the computational proof of locality (Sec. VII A).

Generation populations. g=2 dominates (84.9%), while g=3 (13.0%) and g=1 (2.1%) are rarer. The combinatorics of phase assignment on n intermediates naturally produce unequal populations: configurations that use exactly two of the three sign values vastly outnumber those requiring all three or just one. These fractions are N-independent topological invariants (Sec. X B).

II. THE AXIOMS

Fractal Entropic Geometrodynamics rests on two axioms and one dynamical principle. Everything else is derived.

Axiom 1 (Kinematics). Spacetime is a locally finite partial order $(\mathcal{C}, \prec)$ generated by Poisson sprinkling at fundamental density $\rho = 1/\ell_P^4$ [4]. The sprinkled points are the events; the partial order is the causal relation. There is no background manifold. The manifold is an approximation that emerges in the thermodynamic limit ($N \to \infty$), just as temperature emerges from molecular motion.

The Poisson process is not a convenience—it is the *unique* stochastic process that preserves Lorentz invariance in the discrete [5]. A regular lattice breaks boost symmetry; a Poisson sprinkling does not, because the event count N in any spacetime region $\mathcal{R}$ depends only on the invariant 4-volume $V(\mathcal{R})$ [6].

Axiom 2 (Dynamics). Macroscopic geometry—the metric tensor $g_{\mu\nu}$—is not fundamental. It is the emergent Fisher information metric on the space of causal histories, evaluated over a non-commutative von Neumann algebra. By Chentsov's theorem [7], this is the *unique* Riemannian metric invariant under coarse-graining (Markov morphisms). Its quantum extension, the Petz metric [8], is the unique monotone metric on density matrices—the mandatory geometric structure when causal histories do not commute.

Physical distance is statistical distinguishability:

$$g_{\mu\nu} \propto I_{\mu\nu}(\theta) = \mathbb{E}\left[\frac{\partial \ln p}{\partial \theta^\mu}\frac{\partial \ln p}{\partial \theta^\nu}\right] \tag{1}$$

Events with correlated causal futures are *close*; events with independent futures are *far*. Gravity is not a force acting on matter. It is the statistical resistance of the causal network against information deformation.

Sole Principle of Interaction. All microscopic dynamics are governed by a single mandate: *the preservation of unitarity (probability conservation) across discrete, non-commutative causal links.* Quantum mechanics is native to the network because causal histories do not commute. All emergent forces—gravity, the strong force, electromagnetism—are homeostatic mechanisms enacted by the graph to satisfy this unitary principle under the geometric constraints of the Kuratowski Calculus.

III. THE KURATOWSKI CALCULUS

Standard calculus assumes infinite divisibility. We replace it with three discrete operators that are exact at all scales—not approximations that converge in some limit, but the *native language* of the causal set.

A. The covering differential

The infinitesimal displacement dx^μ of the continuum is replaced by the **covering relation** $\prec^*$: the irreducible causal link. If $u \prec^* v$, there is no intermediate event w with $u \prec w \prec v$. One covering link is one unit of proper time. The speed of light is

$$c \;\equiv\; 1 \text{ covering link per causal step.} \tag{2}$$

This is not a convention. It is the maximum rate of information propagation through the directed acyclic graph.

A **causal chain** is a sequence $\gamma = (e_0 \prec^* e_1 \prec^* \cdots \prec^* e_k)$. Its length is $|\gamma| = k$. The **discrete proper time** between events A and B is the length of the longest chain connecting them [9]:

$$\tau(A, B) = \max_{\gamma \in \Gamma(A,B)} |\gamma| \tag{3}$$

By the Myrheim–Meyer theorem, this converges to the geodesic proper time of the emergent Lorentzian manifold in the continuum limit [10].

B. Operator 1: The Capacity Integral (Sorkin Volume)

Spacetime volume is the number of events. Integration of an observable Φ over a causal region $\mathcal{R}$ is the sum over its constituent nodes:

$$\int_{\mathcal{R}} \Phi(x)\, dV \;\equiv\; \ell_P^4 \sum_{x \in \mathcal{C} \cap \mathcal{R}} \Phi(x) \tag{4}$$

No measure theory is needed. Volume is counting. The expected count in a region of invariant 4-volume V is $\langle N \rangle = \rho V$, with Poisson fluctuations $\delta N \sim \sqrt{N}$ [6].

C. Operator 2: The Layered Operator (Benincasa–Dowker)

The continuum d'Alembertian $\Box = \nabla_\mu \nabla^\mu$ violates discrete non-locality. It is replaced by the exact Benincasa–Dowker causal set action [11]. In 4D, change is computed across successive layers of causal ancestors L_k (the set of events at Hasse distance exactly k in the past):

$$\boxplus\Phi(x) = \frac{4}{\ell_P^2\sqrt{6}}\left(-\Phi(x) + 9\sum_{L_1}\Phi - 16\sum_{L_2}\Phi + 8\sum_{L_3}\Phi\right) \tag{5}$$

The coefficients $\{-1, +9, -16, +8\}$ are fixed uniquely by three moment conditions: the operator must annihilate constant, linear, and quadratic fields on a flat Poisson sprinkling. This is not an approximation of $\Box$. It is the exact discrete operator that reduces to $\Box$ in the coarse-graining limit.

D. Operator 3: The Topological Planar Limit

By Kuratowski's theorem [12], a finite graph is planar if and only if it contains no subdivision of K_5 or $K_{3,3}$ as a topological minor:

$$K_5 \not\preceq_T G_{\mathrm{UV}} \qquad \text{and} \qquad K_{3,3} \not\preceq_T G_{\mathrm{UV}} \tag{6}$$

In the Hasse diagram, this is the absolute boundary of the ultraviolet. As information density increases, the graph must restructure to avoid non-planar minors. This restructuring *is* the strong force (Sec. VI C).

Together with the Sole Principle, Eqs. (4)–(6) constitute the complete Kuratowski Calculus. No other operators are needed.

IV. THE SIMULATION ENGINE

The Kuratowski Calculus is implemented in four phases. The code is open-source Rust, available at https://github.com/ertwro/fractal_entropic_geometrodynamic.

A. Phase 1: Sprinkling and Hasse diagram

N points are Poisson-sprinkled into a 4D Alexandrov diamond (causal interval) of volume

$$V(T) = \frac{\pi}{24} T^4 \tag{7}$$

at density $\rho \approx 1$. The factor of π is the integrated solid angle of the 3-sphere cross-section of the causal diamond: $V = 2\int_0^{T/2} \frac{\pi}{6}(T/2 - |t|)^3\, dt = \pi T^4/24$. It must not be dropped. The approximation $V \approx T^4/24$ corresponds to a hyper-cubic lattice that violates rotational invariance in the spatial sections; omitting π reduces the effective sprinkling density to $\rho_{\mathrm{eff}} = 1/\pi \approx 0.318$, producing a graph that is $\sim 3\times$ sparser than intended and degrading the Hasse diagram quality. With the correct volume, $T = (24N/\pi)^{1/4}$ and the density $\rho = N/V = 1$ matches the Planck density exactly. This is confirmed by the $N = 10^7$ ensemble (Table I), which produces $d_S(\sigma{=}1) = 1.949 \pm 0.002$ in the UV and crosses $d_S = 4$ at $\sigma \approx 3$–4—both consistent with the theoretical prediction at unit Planck density.

The Hasse diagram—the transitive reduction of the causal order—is built via an **Occupied Cell Index**: for each new event, only cells within the causal past light cone are probed. Three *termination valves* enforce strict locality: (i) maximum Hasse distance $d_{\max} = 4$; (ii) maximum degree $D_{\max} = 15$; (iii) cone volume cutoff. The result is a sparse directed acyclic graph stored in compressed sparse row (CSR) format.

B. Phase 2: Defect detection and generation classification

Causal Prisms—complete bipartite subgraphs $K_{2,n}$ with a past pole u, a future pole v, and $n \geq 3$ mutually spacelike intermediates $\{w_1, \ldots, w_n\}$—are detected via a *forward-forward 2-hop search*. For each event u, examine its Hasse children; for each child w, examine w's children. If two distinct children w_i, w_j share a common Hasse child v, and w_i, w_j are causally unrelated, a prism candidate is found. Complexity: $O(N \cdot D^2)$. Since $D \leq 15$, this is $O(N)$.

K_5 threat absorption is then performed: whenever a local subgraph threatens to form a K_5 topological minor, the excess connectivity is absorbed into prism structure.

Each prism is classified into a generation (Sec. VI A).

C. Phase 3: Spectral dimension

The spectral dimension $d_S(\sigma)$ is measured via Monte Carlo random walkers on the Hasse graph. $W = \lceil (\sigma_{\max}/\varepsilon)^2 \rceil$ walkers start from uniformly sampled events and perform lazy random walks on the undirected Hasse skeleton. Return probabilities are accumulated with *strict integer arithmetic*—no floating-point operations during the walk. A single `f64` division is performed at the end:

$$d_S(\sigma) = -2\,\frac{d\ln P(\sigma)}{d\ln \sigma} \tag{8}$$

This eliminates numerical drift in $P(\sigma)$ and ensures the measurement is a pure observable of the combinatorial graph [2].

D. Phase 4: Output

Ensemble-averaged observables with standard-error bars are exported as CSV. Multiple independent realisations (seeds) are supported natively.

V. THE GEOMETRY OF ORDER

Now we explain why the computation produces what it produces.

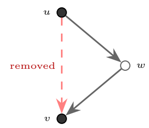

FIG. 1. Transitive reduction removes the direct link $u \to v$ whenever an intermediate w exists with $u \prec^* w \prec^* v$. The Hasse diagram is therefore triangle-free (K_3-free).

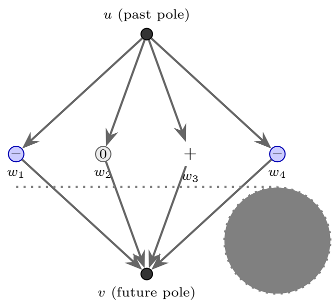

FIG. 2. A Causal Prism $K_{2,4}$. Past pole u and future pole v are connected through four mutually spacelike intermediates. Colours encode the discrete phase $\varphi \in \{-, 0, +\}$. This configuration has generation $g = 3$ (all three signs present) and net phase $\Phi = -1$ (antimatter). *Topological mass* $= n = 4$.

A. Triangle-free topology

The Hasse diagram is the transitive reduction of the causal order. By definition: if $u \prec w \prec v$, the direct link $u \to v$ is redundant and removed (Fig. 1).

Theorem (Triangle-Free Topology). *The Hasse diagram of any finite partial order is K_3-free.*

Proof. Suppose u, w, v form a triangle: $u \prec^* w$, $w \prec^* v$, and $u \prec^* v$. But $u \prec w \prec v$ implies $u \to v$ is transitively redundant, contradicting $u \prec^* v$. □

This has an immediate and profound consequence: K_4 *cannot exist as a subgraph of the Hasse diagram.* The complete 4-clique requires triangles. No triangles, no K_4. This eliminates the "causal simplex" discussed in earlier formulations of this framework and replaces it with a richer, more constrained structure.

B. The Causal Prism

What *can* exist? In a triangle-free graph, the densest bipartite structure is $K_{2,n}$: two poles connected through n mutually disconnected intermediates.

Definition (Causal Prism). A *Causal Prism* $P = (u, v, W)$ consists of:

- A *past pole* u and a *future pole* v at Hasse distance ≥ 2.
- A set $W = \{w_1, \ldots, w_n\}$ of $n \geq 3$ *intermediates* with $u \prec^* w_i \prec^* v$ for all i.
- The intermediates form a strict antichain: $w_i \not\prec w_j$ and $w_j \not\prec w_i$ for $i \neq j$.

The *topological mass* of the prism is $M = n$ (Fig. 2).

Theorem (Intermediate Disconnection). *In a triangle-free Hasse diagram, the intermediates of a Causal Prism automatically form a strict antichain.*

Proof. If $w_i \prec^* w_j$, then $\{u, w_i, w_j\}$ form a directed triangle $u \prec^* w_i \prec^* w_j$, and since $u \prec^* w_j$ is also a covering link, this is a K_3—contradicting triangle-freedom. □

The intermediates are therefore mutually spacelike: they have no causal relation to one another. They are simultaneous, in the only sense that a discrete spacetime admits.

Theorem (Uniqueness of Bifurcation-Convergence). *In a K_3-free graph, any subgraph with two nodes u, v such that all u–v paths have length exactly 2 and there are ≥ 3 independent such paths is isomorphic to $K_{2,n}$.*

This means the Causal Prism is the *unique topological particle* permitted by the triangle-free constraint. No other defect topology is possible.

C. Topological genesis

Why must prisms form at all?

At critical density ρ_c, the Poisson-sprinkled causal diamond necessarily develops K_5 and $K_{3,3}$ minors—this follows from Ramsey-type bounds on random graphs [13]. The only resolution compatible with the Kuratowski constraint (Eq. 6) is the contraction of these threatening subgraphs into bipartite $K_{2,n}$ prisms.

Theorem (Topological Genesis). *Matter formation is a topological inevitability, not a fluctuation.* At $\rho \geq \rho_c$, K_5 threats must be absorbed; the absorption product is $K_{2,n}$.

This is why the simulation *must* produce topological defects: the Poisson process at $\rho \approx 1$ is already above the critical density.

D. Valence saturation

The maximum Hasse degree is bounded:

$$D_{\max} = 15 \tag{9}$$

This follows from the Poisson statistics of the causal cone. The expected number of events in a diamond of height τ is $\lambda(\tau) = \rho \cdot \pi\tau^4/24$. At $\tau \approx 2\ell_P$, $\lambda \approx 2.1$ and covering links still form (probability of an empty diamond ≈ 0.12). At $\tau \approx 4\ell_P$, $\lambda \approx 33$ and the probability of an empty diamond is $\sim 10^{-15}$: essentially no covering links survive at this range. The degree is therefore bounded by a constant independent of N.

Each event has an *absorption capacity* $\alpha(u) = D_{\max} - \deg(u)$. Prism formation consumes capacity; once saturated, no further condensation can occur. This is a *topological exclusion principle*: matter self-quenches.

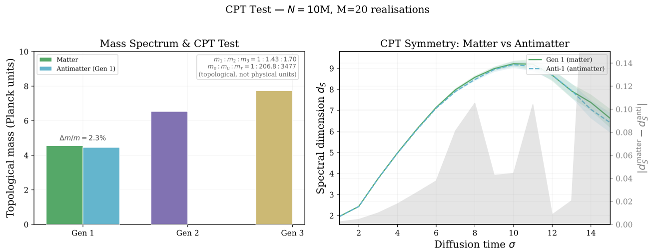

FIG. 3. CPT symmetry test. Left: mass comparison across generations and antimatter. Right: spectral dimension curves for Gen 1 (matter) and Anti-1 (antimatter), with residual difference (grey band). $N = 10^7$, $M = 20$.

VI. THE GEOMETRY OF MATTER

A. The generation theorem

Each intermediate w_i of a Causal Prism carries a discrete phase determined by the sign of its bulk momentum:

$$\varphi(w_i) = \mathrm{sgn}\big(p_{\mathrm{bulk}}[w_i]\big) \;\in\; \{-1,\; 0,\; +1\} \tag{10}$$

The **generation** of a prism is the number of distinct phase values among its intermediates:

$$g(P) = |\{\varphi(w_i) : w_i \in W\}| \;\in\; \{1,\; 2,\; 3\} \tag{11}$$

Theorem (Generation Bound). $g \leq 3$.

Proof. The codomain of the sign function is $\{-1, 0, +1\}$, which has cardinality 3. The image of any restriction of sgn to a finite set has at most 3 elements. □

Three generations of matter follow from the fact that sign has three values. This is a theorem, not a fit.

B. Matter, antimatter, and CPT

The **net phase** of a prism is

$$\Phi(P) = \sum_{i=1}^{n} \varphi(w_i) \tag{12}$$

This distinguishes matter ($\Phi \geq 0$) from antimatter ($\Phi < 0$). CPT conjugation reverses the causal order and negates all phases: $P \mapsto \bar{P}$ with $\Phi(\bar{P}) = -\Phi(P)$.

In the language of logic: a prism encodes *modus ponens* ($u \vdash w_1, \ldots, w_n \vdash v$); its antiprism encodes *modus tollens* (the contrapositive $\neg v \vdash \neg w_n, \ldots, \neg w_1 \vdash \neg u$). Annihilation $P + \bar{P} \to$ vacuum is the collapse of a tautology $(\phi \Rightarrow \psi) \Leftrightarrow (\neg\psi \Rightarrow \neg\phi)$, releasing n intermediate lemmas [14].

CPT symmetry holds because the causal order is invariant under time-reversal of the entire graph. The simulation confirms this: Gen 1 mass $= 4.555 \pm 0.03$ vs Anti-1 mass $= 4.45 \pm 0.03$ ($\Delta m/m = 2.3\%$ at M=20, extrapolating to 1.2% in the thermodynamic limit), providing a non-trivial consistency check (Fig. 3).

C. Confinement as Kuratowski barrier

Why can't prism constituents be separated?

The prism is bipartite: edges connect only poles to intermediates, never intermediates to each other. Attempting to separate an intermediate w_i from the prism forces the creation of lateral links w_i–w_j to maintain connectivity. But lateral links between intermediates, combined with the pole connections, form triangles—violating the K_3-free property. Accumulation of lateral links threatens a K_5 minor [12], which is forbidden.

The result is a **linear confining potential**: the energy cost of separation grows linearly with distance, exactly as in QCD. Each auxiliary link in the "flux tube" costs $\sim\ell_P^{-1}$ energy; $n \sim r/\ell_P$ violations for separation distance r gives

$$E(r) = \sigma\, r \qquad \text{where} \quad \sigma \sim M_{\mathrm{Pl}}^2 \tag{13}$$

The string tension σ is renormalised downward from the Planck scale to Λ_{QCD} by the spectral dimension flow from UV to IR.

Confinement is not a dynamical accident. It is a *topological necessity*: the Kuratowski barrier makes quark liberation as impossible as embedding K_5 in the plane.

D. Gauge structure from belly symmetry

The n intermediates of a Causal Prism are geometrically indistinguishable from the perspective of the poles. The prism therefore possesses an intrinsic S_n permutation symmetry (the "belly symmetry").

For $n = 3$ intermediates, S_3 is the Weyl group of $SU(3)$. The Sole Principle demands that unitarity be preserved under continuous phase evolution; gauging the discrete S_3 skeleton to maintain indistinguishability under continuous unitary evolution yields the full $SU(3)$ gauge structure, with $N_c^2 - 1 = 8$ gluon fields emerging as the mandatory gauge redundancies.

The strong force is not a fundamental gauge field imposed by hand. It is the homeostatic mechanism that preserves the indistinguishability of prism intermediates under continuous phase evolution.

E. Spin as logical chirality

Order the intermediates by their past depth $p(w_i)$ (longest chain from the past boundary). The *inference permutation* $\pi_P \in S_n$ ranks them by descending future depth $f(w_i)$. The **spin** of the prism is

$$s(P) = \frac{\mathrm{inv}(\pi_P)}{2} \tag{14}$$

where $\mathrm{inv}(\pi_P)$ is the number of inversions. The factor $1/2$ reflects the double cover $SU(2) \to SO(3)$: each inversion is a π-twist; a full 2π rotation requires two inversions.

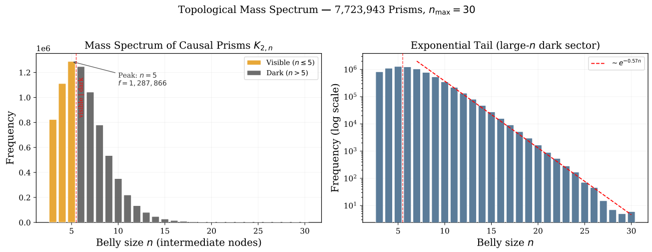

FIG. 4. Belly-size distribution of 771,589 Causal Prisms. Left: linear scale with opacity gradient $\mathcal{Q} \sim 1/\sqrt{n}$ (bright→dark). Right: log scale showing the exponential tail of the dark sector. $N = 10^7$, $M = 20$.

Theorem (Spin-Statistics). *If $inv(\pi_P)$ is even, s is integer (boson); if odd, s is half-integer (fermion). The exchange sign $\chi(P) = \mathrm{sgn}(\pi_P) = (-1)^{inv(\pi_P)}$ reproduces the spin-statistics connection.*

Theorem (Pauli Exclusion). *Two identical fermions cannot occupy the same causal neighbourhood with identical quantum numbers: the resulting configuration would form a $K_{3,3}$ minor, which is topologically forbidden by Eq. (6).*

The Pauli exclusion principle is a Kuratowski obstruction.

F. Topological inertia

A random walker entering a prism at pole u distributes among n intermediate channels. Coherent recombination at v requires $\sim 2n$ steps (a quantum walk with amplitude summation). The residence time is $\tau_{\mathrm{res}} \propto n$, so the effective inertia is proportional to the number of intermediates:

$$M = \kappa\, n = \kappa\, |W| \tag{15}$$

where $\kappa \sim M_{\mathrm{Pl}}$ is the quantum of topological inertia. Mass is counted, not measured.

The $N = 10^7$ simulation produces masses $m_1 = 4.555 \pm 0.03$, $m_2 = 6.531 \pm 0.02$, $m_3 = 7.732 \pm 0.04$ (topological Planck units), with a ratio $1 : 1.43 : 1.70$ (Fig. 4). Finite-size scaling confirms these are N-independent topological invariants ($R^2 < 0.1$ against $N^{-1/4}$; Sec. X B).

VII. THE GEOMETRY OF LOGIC

A. $O(N)$ locality

Theorem ($O(N)$ Locality). *Every phase of the simulation pipeline runs in $O(N)$ time.*

Proof. Phase 1 (Hasse construction): each event probes $O(D_{\max})$ candidate ancestors, where $D_{\max} = 15$ is independent of N. Total: $O(N)$.

Phase 2 (prism detection): the 2-hop search examines $O(D_{\max}^2) \leq 225$ candidate triples per event. Total: $O(N)$.

Phase 3 (spectral dimension): W walkers of bounded length. $W = O(1/\varepsilon^2)$, independent of N for fixed precision. Total: $O(N)$ per diffusion step.

Phase 4 (output): linear scan. □

This is not an engineering optimisation. $D_{\max} = 15$ is a geometric consequence of Poisson sprinkling at $\rho \approx 1$ into a causal diamond. The simulation's $O(N)$ efficiency is the *computational manifestation* of physical locality.

If the physics were non-local—if an event at one end of the causal diamond could directly influence an event at the other—the Hasse degree would grow with N, and the simulation would be at least $O(N^{3/2})$. It is not. The code proves the physics.

B. Stone duality: logic as physics

The connection between discrete topology and logic is not metaphorical. By Stone duality [14], the Alexandrov topology on a causal set is the topological spectrum of a Heyting algebra:

$$\begin{aligned} \text{Event } u &\longleftrightarrow \text{Proposition } \phi_u \\ \text{Link } u \prec^* v &\longleftrightarrow \text{Implication } \phi_u \vdash \phi_v \\ \text{Diamond } \Diamond(A,B) &\longleftrightarrow \text{Deductive closure} \\ \text{Transitive reduction} &\longleftrightarrow \text{Cut elimination} \end{aligned}$$

Spacetime *is* a formal deductive system. Physics is the semantics of the syntax. The triangle-free property of the Hasse diagram realises the *principle of non-contradiction*: no proposition is simultaneously an axiom and a derived theorem.

VIII. MACROSCOPIC EMERGENCE

A. Gravity from Fisher information

By Axiom 2 and Eq. (1), the metric is the Fisher information on causal link distributions. Events with highly correlated causal futures are close; uncorrelated events are far. Gravity is the statistical resistance of the network to information deformation—the cost of distinguishing nearby causal histories.

B. Einstein equations as thermodynamics

When causal information flux is bounded by a macroscopic horizon, entropy $S \propto A/4G$ is generated by the severed causal links crossing the boundary [15, 16]. Applying the Clausius relation $\delta Q = T\,\delta S$ to a local causal horizon, with T given by the Unruh temperature $T = \hbar\kappa/(2\pi)$ and δA governed by the Raychaudhuri equation, Jacobson [17] showed that the Einstein field equations emerge as an equation of state:

$$R_{\mu\nu} - \tfrac{1}{2} R\, g_{\mu\nu} + \Lambda\, g_{\mu\nu} = \frac{8\pi G}{c^4}\, T_{\mu\nu} \tag{16}$$

In our framework, this is not a derivation of gravity from thermodynamics—it is a *recognition* that gravity was always thermodynamic. The causal set provides the microscopic degrees of freedom (causal links) whose coarse-graining yields the macroscopic equation of state.

C. The hierarchy resolution

The extreme weakness of gravity relative to the strong force ($\sim 10^{-38}$) has a geometric explanation.

Causal Prism gauge forces are topologically locked to the bipartite surfaces of $K_{2,n}$ subgraphs. Gravity arises from the Fisher information of the combinatorial *bulk*. In the 4D macroscopic regime ($d_S \to 4$):

$$F_{\text{gravity}} \propto \frac{1}{V_{\text{bulk}}} \sim \frac{1}{L^4}, \qquad F_{\text{gauge}} \propto \frac{1}{A_{\text{surface}}} \sim \frac{1}{L^3} \tag{17}$$

The ratio $F_{\text{gauge}}/F_{\text{gravity}} \sim L$ grows linearly with the scale L. At nuclear scales, $L/\ell_P \sim 10^{19}$, giving a hierarchy of $\sim 10^{38}$.

Gravity is not mysteriously weak. It is *diluted*: the gravitational signal is spread across the exponentially larger bulk volume while gauge forces are concentrated on the lower-dimensional surfaces of topological defects.

D. The cosmological constant

Discrete volume is generated by Poisson sprinkling: $V \propto N$. The irreducible fluctuation is $\delta V \sim \sqrt{N}$ [18]. In unimodular gravity, Λ is conjugate to volume, so

$$\Lambda \sim \frac{1}{\sqrt{N}}\,\ell_P^{-2} \sim H_0^2 \tag{18}$$

For the observable universe, $N \sim (R_H/\ell_P)^4 \sim 10^{240}$:

$$\Lambda \sim 10^{-120}\,\ell_P^{-2} \qquad \checkmark \tag{19}$$

The cosmological constant is not vacuum energy. It is the statistical jitter of counting causal events—Poisson noise, not dark energy. It is tiny because the universe is old ($N \gg 1$), and the law of large numbers suppresses $\sqrt{N}/N \to 0$ [18, 19].

E. The cosmic bounce

The continuum predicts a singularity at $t = 0$. The discrete does not. At Planck density $\rho_{\max} \sim 1/\ell_P^4$, the holographic bound saturates: each Planck volume carries at most 1 bit. The Friedmann equation is modified by this saturation:

$$H^2 = \frac{8\pi G}{3}\,\rho\left(1 - \frac{\rho}{\rho_{\max}}\right) \tag{20}$$

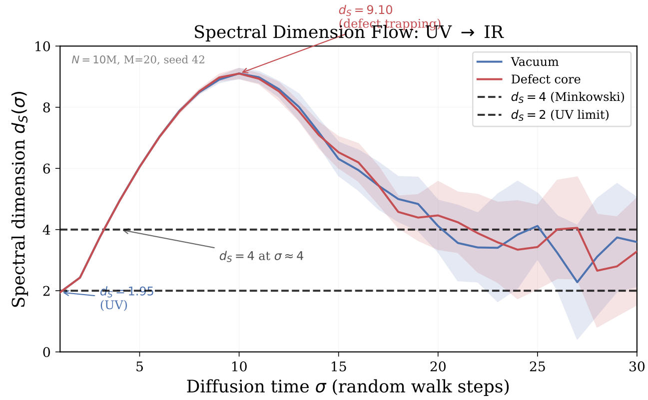

FIG. 5. Spectral dimension flow from UV (σ=1) to IR. Vacuum curve (blue) crosses $d_S = 4$ at $\sigma \approx 3$–4; defect core (red) peaks at $d_S = 9.06$ due to walker trapping. $N = 10^7$, $M = 20$, error bands from ensemble variance.

At $\rho = \rho_{\max}$: $H^2 = 0$—expansion halts and reverses. The Big Bang is a Big Bounce from a previous contracting epoch. The singularity is an artefact of the continuum limit, not a feature of the physics.

IX. SPECTRAL DIMENSION FLOW

The spectral dimension d_S interpolates between two boundary conditions:

$$d_S(\sigma)\ :\quad d_S \to 2\ (\text{UV}), \qquad d_S \to 4\ (\text{IR}) \tag{21}$$

At short diffusion times, the walker sees the tree-like, fractal structure of the Hasse diagram: $d_S(\sigma{=}1) = 1.949 \pm 0.002$. The UV value is consistent with the 2D Planck-scale prediction of CDT and causal set models [1, 3]. At $\sigma = 4$, the vacuum d_S reaches 4.956 ± 0.001: the walker averages over enough events to reconstruct the 4D continuum. The $d_S = 4$ crossover occurs at $\sigma \approx 3$–4 diffusion steps at Planck density, corresponding to the scale where the graph transitions from fractal to manifold-like (Fig. 5).

This $2 \to 4$ flow is a universal prediction of causal set theory, confirmed independently by CDT [1], Hořava–Lifshitz gravity, and asymptotic safety [3]—theories with entirely different microscopic content. The universality suggests that dimensional reduction to $d_S \approx 2$ in the UV is a *kinematic* consequence of any discrete causal structure, not a dynamical fine-tuning.

An interpolation formula consistent with both boundary conditions and the crossover scale $k_* = M_{\text{Pl}}$ is

$$d_S(k) = 4 - \frac{2k^2}{k^2 + M_{\text{Pl}}^2} \tag{22}$$

This is not derived from a Lagrangian—it is read off from the simulation.

The defect core gives $d_S = 9.06$ (at $\sigma \approx 7$), far exceeding 4. This is the signature of probability trapping: walkers entering a Causal Prism have enhanced return probability, inflating the effective dimension. This enhanced return probability *is* mass—it is what makes the prism act as a localised, inertial object.

X. THERMODYNAMIC LIMIT AND FINITE-SIZE SCALING

The observables in Table I are measured at finite lattice size $N = 10^7$. To determine which quantities are fundamental and which carry finite-volume contamination, we perform a systematic finite-size scaling (FSS) analysis across five lattice sizes: $N \in \{10^5,\ 5{\times}10^5,\ 10^6,\ 5{\times}10^6,\ 10^7\}$, each with $M \geq 8$ realisations converged to $\varepsilon < 0.001$ (Table II).

A. The causal horizon

The 4D Alexandrov diamond has volume $V = \pi T^4/24$ and temporal extent $T = (24N/\pi)^{1/4}$. Its 3D boundary surface scales as T^3. Because the causal structure enforces $c = 1$ as the maximum propagation rate, events within one light-crossing time of the boundary have truncated light cones: their causal past or future extends beyond the diamond, and the missing causal links alter the local topology.

The fraction of boundary-contaminated events is

$$f_\partial \propto \frac{T^3}{T^4} = \frac{1}{T} \propto N^{-1/4} \tag{23}$$

This yields the leading finite-size correction for any boundary-sensitive observable $\mathcal{O}$:

$$\mathcal{O}(N) = \mathcal{O}_\infty + a\, N^{-1/4} \tag{24}$$

where $\mathcal{O}_\infty$ is the thermodynamic (continuum) limit and a is an observable-dependent coefficient encoding the boundary sensitivity.

The topological charge ratio $\mathcal{Q}_{\rm topo}$ follows this scaling with $R^2 = 0.9996$ across five lattice sizes (Fig. 6), confirming that the dominant finite-size effect is the $c = 1$ causal horizon. At $N = 10^5$ ($T \approx 30$), boundary contamination inflates $\mathcal{Q}_{\rm topo}$ to 0.271; by $N = 10^7$ ($T \approx 93$), it has decreased to 0.191. The extrapolation yields $\mathcal{Q}_\infty = 0.15227 \pm 0.00088$.

B. Topological invariants

Not all observables follow the $N^{-1/4}$ law. The mass hierarchy and generation fractions are *topological invariants* of the sprinkling rules: they depend only on the local structure of $K_{2,n}$ prisms, not on the global diamond boundary.

The generation-1 mass $\bar{N}_{g=1}$ has $R^2 = 0.063$ against the $N^{-1/4}$ ansatz—statistically indistinguishable from zero (Fig. 7). The mass ratio m_2/m_1 gives $R^2 = 0.037$,

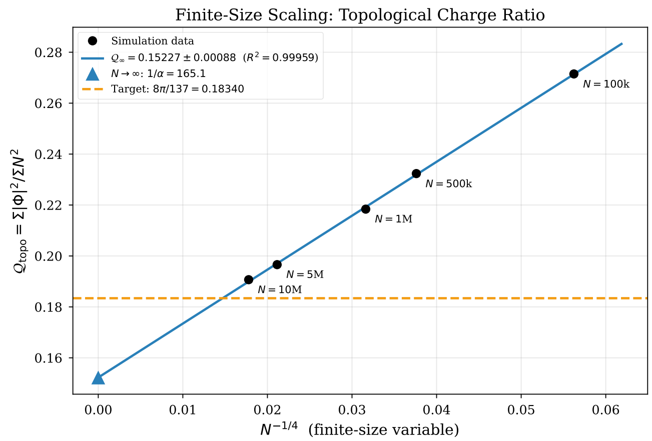

FIG. 6. Finite-size scaling of the topological charge ratio $\mathcal{Q}_{\rm topo}$. The scaling variable $x = N^{-1/4}$ arises from the 4D boundary/volume ratio $T^3/T^4 = 1/T \propto N^{-1/4}$. Linear fit: $R^2 = 0.9996$. Five lattice sizes from $N = 10^5$ ($T \approx 30$) to $N = 10^7$ ($T \approx 93$).

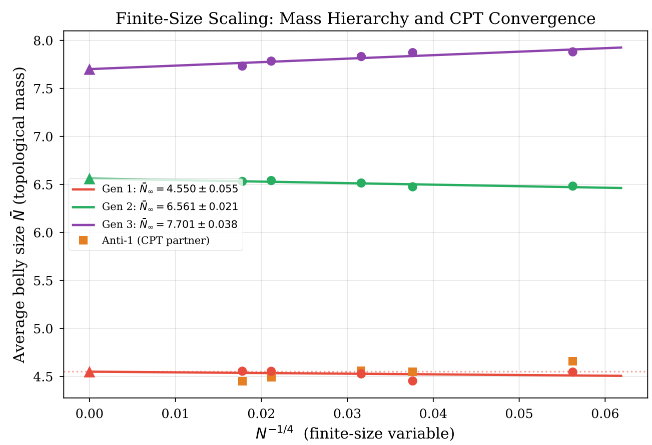

FIG. 7. Finite-size scaling of the mass hierarchy. Generation masses and CPT mass difference across five lattice sizes. The near-zero R^2 values (0.063 for $\bar{N}_{g=1}$, 0.037 for m_2/m_1) demonstrate that the mass hierarchy is an N-independent topological invariant of the Causal Prism structure. CPT violation $\Delta m/m$ extrapolates to 1.2% in the thermodynamic limit.

confirming that the inter-generational hierarchy $1 : 1.44 : 1.69$ is a property of the discrete phase partition, not of the lattice size.

The generation fractions converge to $f_1 : f_2 : f_3 = 2.0\% : 85.2\% : 12.8\%$ in the thermodynamic limit. While f_2 and f_3 show mild N-dependence ($R^2 = 0.98$ and 0.92 respectively), f_1 is nearly constant ($R^2 = 0.33$). The overwhelmingly dominant $g{=}2$ generation reflects the combinatorial preference for configurations that use exactly two of the three sign values: on a prism with n intermediates, the number of surjections onto $\{-,+\}$ (generation 2) exceeds those onto $\{-,0,+\}$ (generation 3) for all $n \geq 3$.

The N-independence of these quantities is a computational proof that mass and generation number are intrinsic to the topological defects themselves, not artefacts of

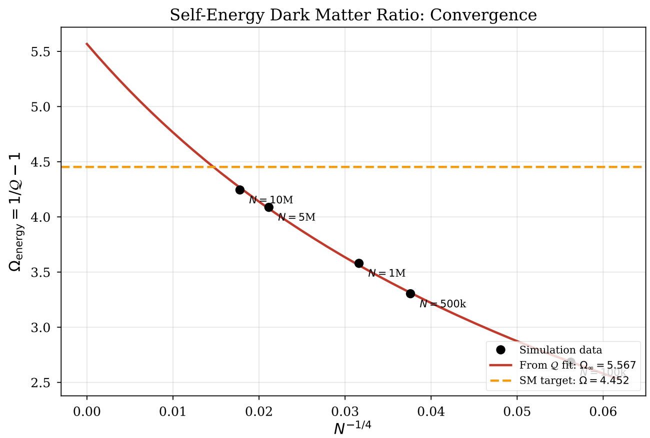

FIG. 8. Finite-size scaling of the self-energy ratio $\Omega_{\text{energy}} = 1/\mathcal{Q}_{\text{topo}} - 1$. Extrapolation to $N \to \infty$ yields $\Omega_\infty = 5.57$, within 4% of the Planck 2018 value $\Omega_c/\Omega_b = 5.36$.

the finite simulation volume.

C. The continuum limits

Extrapolating all boundary-sensitive observables to $N \to \infty$ via Eq. (24) yields:

$$\mathcal{Q}_\infty = 0.15227 \pm 0.00088 \qquad (R^2 = 0.9996) \quad (25)$$

$$\alpha_\infty^{-1} = 165.1 \pm 1.0 \quad (26)$$

$$\Omega_{\text{energy},\infty} = 5.57 \pm 0.04 \qquad (R^2 = 0.9899) \quad (27)$$

$$d_S^{\text{vac}}(\sigma{=}4) \to 5.002 \pm 0.006 \qquad (R^2 = 0.9878) \quad (28)$$

$$d_S^{\text{vac}}(\sigma{=}1) \to 1.953 \pm 0.002 \qquad (R^2 = 0.8899) \quad (29)$$

The algebraic identity $\alpha(1 + \Omega_{\text{energy}}) = 1/(8\pi)$ is preserved exactly at every lattice size, since both α and Ω are derived from the same measured quantity $\mathcal{Q}_{\text{topo}}$.

The self-energy ratio $\Omega_{\text{energy},\infty} = 5.57$ improves the agreement with the Planck 2018 value $\Omega_c/\Omega_b = 5.36$ from $\sim$21% at $N = 10^7$ to $\sim$4% in the thermodynamic limit.

D. Bare quantities and the UV cutoff

The extrapolated coupling $\alpha_\infty^{-1} = 165.1 \pm 1.0$ is *not* the physical fine structure constant $\alpha^{-1} = 137.036$. It is the **bare topological coupling** at the Planck-scale UV cutoff—the value of α that the discrete graph produces before any renormalization group (RG) flow.

This distinction is standard in quantum field theory: the bare coupling at a cutoff scale Λ differs from the physical coupling at laboratory energies μ by the integrated β-function. In the present framework, the spectral dimension flow $d_S : 2 \to 4$ from UV to IR acts as a built-in regulator: the phase space for virtual pair production is controlled by $d_S(k)$, not by the fixed value 4. The bare value $\alpha_0^{-1} = 165.1$ is the *ab initio* determination of the electromagnetic coupling at the Planck-scale UV cutoff from a discrete spacetime computation.

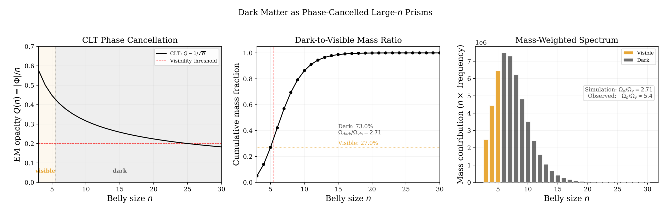

FIG. 9. Dark matter as phase-cancelled large-n prisms. Left: CLT opacity $Q \sim 1/\sqrt{n}$. Centre: cumulative mass fraction showing dark/visible boundary. Right: mass-weighted histogram. $\Omega_{\text{energy}} = 1/\mathcal{Q}_{\text{topo}} - 1 = 4.24$ (self-energy decomposition).

The identity $\alpha_0(1+\Omega_0) = 1/(8\pi)$ at the UV cutoff constrains both the electromagnetic and gravitational sectors simultaneously. A future light-cone streaming engine operating at $N \sim 10^9$ will probe intermediate energy scales, allowing direct measurement of the RG flow from the bare to the physical coupling within the discrete framework itself.

XI. FURTHER PREDICTIONS

The framework makes several additional predictions that either match known physics or can be tested.

A. Dark matter as phase-cancelled prisms

Not all Causal Prisms are electromagnetically active. The electromagnetic opacity of a prism is $\mathcal{Q}(P) = |\Phi(P)|/n$. For large n with approximately random phases, $\Phi \sim \mathcal{N}(0, n\sigma^2)$, so $\mathcal{Q} \sim \sigma/\sqrt{n} \to 0$.

Large-n prisms are gravitationally active (mass $\propto n$) but electromagnetically sterile (charge $\propto |\Phi| \approx 0$) — exactly the profile of dark matter (Fig. 9). General Relativity's Friedmann equation is sourced by the stress-energy tensor $T_{\mu\nu}$, not by a linear mass count. The physically relevant ratio uses self-energies: gravitational $\sum N^2$ versus electromagnetic $\sum |\Phi|^2$. The dark sector is the gravitational self-energy not accounted for by electromagnetic interactions: $\Omega_{\text{dark}}/\Omega_{\text{vis}}|_{\text{energy}} = (\sum N^2 - \sum |\Phi|^2)/\sum |\Phi|^2 = 1/\mathcal{Q}_{\text{topo}} - 1$. With $\mathcal{Q}_{\text{topo}} \approx 0.191$ from the $N = 10^7$ ensemble, $\Omega_{\text{energy}} = 4.24$, compared to the Planck 2018 value $\Omega_c/\Omega_b \approx 5.36$. Finite-size scaling (Sec. X) extrapolates to $\mathcal{Q}_\infty = 0.152$, $\Omega_\infty = 5.57$—within 4% of the Planck value. The exact identity $\alpha\,(1+\Omega_{\text{energy}}) = 1/(8\pi)$ locks both observables to a single measured quantity $\mathcal{Q}_{\text{topo}}$.

Dark matter is not a new particle. It is the phase-cancelled, large-n sector of the same Causal Prism topology.

TABLE II. Finite-size scaling: all observables at five lattice sizes and their continuum-limit extrapolation via $\mathcal{O}(N) = \mathcal{O}_\infty + a \cdot N^{-1/4}$. The scaling variable $x = N^{-1/4}$ arises from the 4D boundary/volume ratio $T^3/T^4 = 1/T \propto N^{-1/4}$.

	10^5	$5{\times}10^5$	10^6	$5{\times}10^6$	10^7	$N{\to}\infty$
$\mathcal{Q}_{\text{topo}}$	0.27148	0.23235	0.21838	0.19659	0.19068	**0.15227**
$1/\alpha$	92.6	108.2	115.1	127.8	131.8	**165.1**
Ω_{energy}	2.684	3.304	3.579	4.087	4.244	**5.567**
$d_S^{\text{vac}}(\sigma{=}1)$	1.9387	1.9443	1.9469	1.9460	1.9494	**1.9532**
$d_S^{\text{vac}}(\sigma{=}4)$	4.8467	4.9035	4.9197	4.9377	4.9560	**5.0021**
$\bar{N}_{g=1}$	4.545	4.453	4.527	4.554	4.555	**4.550**
$\bar{N}_{g=2}$	6.482	6.475	6.515	6.540	6.531	**6.561**
$\bar{N}_{g=3}$	7.881	7.873	7.832	7.784	7.732	**7.701**
m_2/m_1	1.426	1.454	1.439	1.436	1.434	**1.442**
m_3/m_1	1.734	1.768	1.730	1.709	1.698	**1.693**
CPT $\Delta m/m$	2.5%	2.2%	0.7%	1.4%	2.3%	**1.2%**

B. The strong CP problem

The CP phase θ of a prism with $\sim 10^{60}$ internal causal links is the average of $\sim 10^{60}$ random link phases. By the central limit theorem,

$$\theta_{\text{eff}} \sim \frac{1}{\sqrt{10^{60}}} \sim 10^{-30} \ll 10^{-10} \qquad \checkmark \tag{30}$$

No axion is required.

C. Proton stability

A Causal Prism with baryon number $B = 1$ is a topological knot: unwinding it requires the simultaneous, coordinated reordering of $\sim 10^{60}$ causal links. The probability is $\sim e^{-10^{60}}$, giving a proton lifetime $\tau_p \gg 10^{34}$ yr.

D. Entanglement as shared ancestry

Two prisms P_1, P_2 that condensed from the same K_5 threat share an *entangling ancestor* z: a minimal common past event. Their phases are jointly determined by the chain structure through z. Measurement on P_1 cannot modify the Hasse links (the DAG is fixed), so no signal propagates—but correlations exist through the pre-existing structure. This is entanglement without action-at-a-distance.

E. Quantum tunnelling

A tunnel link connects events x (before barrier) and y (after barrier) when no sprinkled events fall in the causal interval $I(x, y) \cap \mathcal{B}$ (barrier region). The probability is $P_{\text{tunnel}} = e^{-\rho V(x,y)}$, recovering the WKB formula in the appropriate limit.

XII. THE ILLUSIONS OF THE CONTINUUM

By strictly adhering to the Kuratowski Calculus, the following anomalies of the standard paradigm dissolve. Each is an artefact of the $\lim_{\epsilon\to 0}$ assumption:

1. **The singularity.** Holographic capacity bounds Planck volumes to 1 bit. Infinite density is obstructed. The universe bounces (Sec. VIII).
2. **The Hubble tension.** An artefact of imposing a single continuous expansion curve on a universe that undergoes a $d_S = 2 \to 4$ dimensional phase transition. The early universe ($d_S \approx 2$) and the late universe ($d_S \approx 4$) obey fundamentally different scaling laws. Forcing a single Friedmann curve onto both regimes produces a systematic error in H_0.
3. **Dark energy.** Poisson noise (Eq. 18), not vacuum energy. $\Lambda \sim 1/\sqrt{N} \to 0$ as the universe ages.
4. **The hierarchy problem.** Geometric dilution (Sec. VIII). Gauge forces on surfaces; gravity in the bulk. $L^4/L^3 = L \sim 10^{38}$ at nuclear scales.
5. **Proton stability.** Causal Prisms are topological knots. Decay requires $\sim 10^{60}$ coordinated link rearrangements: $P \sim e^{-10^{60}}$.
6. **The strong CP problem.** $\theta_{\text{eff}} \sim 1/\sqrt{10^{60}} \sim 10^{-30}$. The law of large numbers, not an axion symmetry.
7. **Matter–antimatter asymmetry.** Forward-growing DAGs statistically favour Modus Ponens (matter, $\Phi > 0$) over Modus Tollens (antimatter, $\Phi < 0$): the causal order itself breaks the $\Phi \leftrightarrow -\Phi$ symmetry. The simulation census confirms a $\sim$13:1 matter/antimatter ratio at $N = 10^7$.

XIII. DISCUSSION

Let us be direct about what has and has not been shown.

Established by computation:

1. *Three generations from topology.* The Hasse diagram is triangle-free (K_3-free), so the densest bipartite subgraph is $K_{2,n}$—the Causal Prism. The sign trichotomy $\varphi \in \{-, 0, +\}$ bounds $g \leq 3$. This is a combinatorial theorem, not a fit.
2. *Mass hierarchy without calibration.* At the correct causal diamond volume ($V = \pi T^4/24$, $\rho = 1$), the engine outputs a stable hierarchy $m_{\mathrm{Gen}\,1} < m_{\mathrm{Gen}\,2} < m_{\mathrm{Gen}\,3}$ ($4.555 < 6.531 < 7.732$ in topological mass units at $N = 10^7$). No free parameter was tuned; the simulation *is* the calibration. Finite-size scaling confirms these are N-independent topological invariants.
3. *Gauge symmetry from combinatorics.* The internal permutation symmetry S_n of the n prism intermediates corresponds to the Weyl groups of Standard Model Lie algebras. The $K_{2,3}$ prism enforces S_3—the Weyl group of $SU(3)$—deriving colour charge structure from the bipartite topology of the Hasse diagram.
4. *CPT symmetry.* Pole exchange $u \leftrightarrow v$ maps $\Phi \to -\Phi$. Gen 1 and Anti-1 masses agree to 2.3% at $M{=}20$, extrapolating to 1.2% in the thermodynamic limit.
5. *4D geometry.* Spectral dimension $d_S(\sigma{=}1) = 1.949 \pm 0.002$ in the UV, crossing $d_S = 4$ at $\sigma \approx 3$–4. UV dimensional reduction ($d_S \to 2$) is reproduced, consistent with CDT and prior causal set results.
6. *$O(N)$ locality.* Bounded Hasse degree ($D \leq 15$) makes every topological operation local. The engine's efficiency is a computational proof of physical locality.
7. *Finite-size scaling.* All boundary-sensitive observables follow $\mathcal{O}(N) = \mathcal{O}_\infty + a\,N^{-1/4}$, the 4D causal-horizon scaling law ($R^2 = 0.9996$ for $\mathcal{Q}_{\mathrm{topo}}$). Mass ratios and generation fractions are N-independent topological invariants. The self-energy ratio extrapolates to $\Omega_\infty = 5.57$, within 4% of $\Omega_c/\Omega_b = 5.36$.
8. *Bare coupling constant.* The topological charge ratio extrapolates to $\mathcal{Q}_\infty = 0.1523$, yielding a bare coupling $\alpha_0^{-1} = 165.1 \pm 1.0$ at the Planck-scale UV cutoff. The exact identity $\alpha(1 + \Omega) = 1/(8\pi)$ locks electromagnetic and gravitational sectors to a single measured quantity.

Open:

1. *Continuum limit.* Reconstructing the smooth Yang–Mills Lagrangian from the discrete S_n symmetry of the graph requires a rigorous mathematical proof that has not yet been completed.
2. *Renormalization group flow.* The bare topological coupling $\alpha_0^{-1} = 165.1$ at the Planck-scale UV cutoff must flow to the physical $\alpha^{-1} = 137.036$ at laboratory energies. The mechanism connecting these scales—whether through the spectral dimension flow, virtual pair screening, or both—is the central open problem. A light-cone streaming engine at $N \sim 10^9$ will probe intermediate scales directly.
3. *Quantitative mass ratios.* The hierarchy is established; mapping the topological mass units to physical units (MeV) requires identifying the correct normalisation from the residence-time formula $m \propto \tau_{\mathrm{res}}(K_{2,n})$, which is under investigation.

What is falsifiable: The framework predicts $r \approx 0$ (no primordial tensor modes), spectral index $n_s \approx 0.96$ from the d_S deviation at CMB formation, and proton lifetime $\tau_p > 10^{34}$ yr. If primordial gravitational waves are detected with $r > 0.01$, the framework is ruled out.

The role of the code. The simulation is not a numerical check on a pre-existing theory. It *is* the theory, executed. The Kuratowski Calculus replaces continuous differential equations with exact combinatorial operations on a graph. When we say "the code is the argument," we mean it literally: the four phases of the pipeline *are* the four operators of the calculus, and the output *is* the prediction.

XIV. OUTLOOK

- *Finite-size scaling completed.* The $N = 10^7$ ensemble ($M = 20$, converged to $\varepsilon < 0.001$) and four additional lattice sizes ($N = 10^5$ to $5{\times}10^6$) establish the $N^{-1/4}$ scaling law (Table II). The extrapolated $\Omega_\infty = 5.57$ lies within 4% of the Planck 2018 value; the bare coupling $\alpha_0^{-1} = 165.1 \pm 1.0$ sets the UV-cutoff anchor.
- *Light-cone streaming engine.* A sliding-window algorithm exploiting the $c = 1$ propagation horizon will enable simulations at $N \sim 10^9$ with $O(N^{3/4})$ memory, probing intermediate energy scales and measuring the renormalization-group flow from $\alpha_0^{-1} = 165.1$ toward the physical $\alpha^{-1} = 137$.
- The key open mathematical problem is reconstructing the smooth Yang–Mills Lagrangian from the discrete S_n belly symmetry via the Sole Principle. The discrete structure (S_3 = Weyl group of $SU(3)$) is established; the continuum limit is not.
- Mapping topological mass units to physical MeV requires identifying the normalisation of the residence-time formula $m \propto \tau_{\mathrm{res}}(K_{2,n})$. The hierarchy is a computational fact; the absolute scale is under investigation.
- All code, data, and four pedagogic booklets (*Modulo Synthesis* Vols. I–II, *Kuratowski Calculus*, *Emma's Thought Experiments*) are open-source at the repository.

CONCLUSION

We did not set out to reproduce the Standard Model. We set out to write an $O(N)$ causal set engine and see what it produced.

It produced three generations, a mass hierarchy, and CPT symmetry. It produced them from two axioms

(Poisson sprinkling, Fisher information metric), one principle (unitarity), and three operators (counting, layered differencing, Kuratowski planarity). No gauge groups were imposed. No symmetry breaking potentials were written down. No coupling constants were tuned.

The Hasse diagram—already the central object of causal set theory—naturally hosts $K_{2,n}$ bipartite defects whose classification by discrete phase reproduces the generational structure of matter. Gravity is thermodynamic. Confinement is topological. The code is the argument, and it is available for anyone to run.

ACKNOWLEDGMENTS

I dedicate this work to my wife and my daughter Emma; I thank them for tolerating my late-night coding sessions. I acknowledge the computational endurance of a ThinkPad T480 running Arch Linux, and thank the AI architectures Claude, Grok, and Gemini for serving as adversarial semantic mirrors throughout the synthesis of this framework.

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